The Future of eBusiness

Scenario

Example – Uber – dynamic pricing using Artificial Intelligence and

Machine Learning In February 2023 Uber announced the introduction of its ‘dynamic pricing’ algorithm to be applied to UK drivers in London to allow the company to set variable pay and pricing levels based on real-time data for market conditions. These market conditions include time, distance, predicted route, estimated traffic and number of users requesting services. This process is now often referred to as ‘surge pricing’.

Uber has a dedicated network within the company structure – Uber AI – it

is a centre for advanced artificial intelligence research and platforms and began utilising research results and outcomes in 2019 when using a mixture of Machine Learning and Artificial Intelligence. It was able to implement new algorithms into the rideshare platform that enabled it to ‘overcome the limits of GPS’ when improving the accuracy of information relating to driver and rider locations and estimated arrival times. The improved service was attributed to developments within the AI Sensing and Perceptions team. Also, in 2019 Uber announced its development of the Plato Research and Dialogue System – a conversational platform powered by AI. Conversational platforms can vary from chatbots to sophisticated digital assistants and Uber’s open-source platform includes functions such as speech recognition, language generation and speech synthesis to convert text to speech.

Uber are committing time and resources to multiple emerging technologies to increase market share, improve customer experience and demonstrate continuous improvement and leadership within their industry.

Brief introduction

From new and emerging technologies to the critical benefits of e-commerce, it is clear that the fast-paced advancement of e-commerce has forever changed the way businesses operate. Organizations are using innovations like artificial intelligence (AI), and automation to reduce operational costs. As an example, Uber uses dynamic pricing algorithms which depend on real market conditions to show how AI can enhance the performance of a business. In a similar manner, Amazon is considered one of the largest e-commerce companies in the world, but never ceased to implement state-of-the-art technologies in order to strengthen its place between digital marketplaces. In this report, we will look at how Amazon uses the emerging technologies to improve its e-commerce business especially in terms of customer relationship, supply chain management, and business Strategies. Through an analysis of Amazon, this report seeks to provide an understanding of the future of e-business and the influence that technology may have on its evolution.

Task 1

Amazon is considered one of the most successful organizations in the world. Amazon has consistently taken the lead in implementing cutting-edge technologies to improve its online business operations.

In the last five years , The company has succeeded in taking its business to the next level by incorporating a number of innovative technologies into its operations such as artificial intelligence (AI), machine learning (ML),automation, and check out free payment systems . It is interesting to know how this company managed to apply these latest technologies to make them one of the most customer-centric businesses in the world in terms of improving customer experience.

One of the most interesting technologies Amazon has adopted is Artificial Intelligence (AI) and Machine Learning (ML). Amazon uses artificial intelligence as a recommendation engine which uses machine learning algorithms to analyze customer behavior, preferences, and purchase history. Amazon was the first enterprise to offer product recommendations on its platform and it has been using it for 20 years. The use of AI allows Amazon to boost the efficiency of its business operations, provide highly personalized product recommendations, enhance shopping experience, and increase sales. For example, when the customer tends to purchase a certain product, the platform automatically suggests relevant items based on the browsing and purchasing activities of the customer.

Another interesting technology that Amazon has adopted is Automation. The company has invested in robotics to optimize its warehouse operations by producing and deploying robotic systems such as Titan, Sparrow, proteus, and sequoia. Titan is a mobile robot whose capabilities are to lift up to 2,500 pounds, this feature can enhance material transportation and ensure its efficiency and safety in its fulfillment centers, furthurmore, sparrow is an AI powered robotic arm that can automates picking up and moving each item from containers to specific totes to send off to employees before packaging. Sparrow uses Computer vision and Artificial intelligence to identify the appropriate item and add it to the tote during the delivery process. In addition, Amazon uses the mobile robots Proteus and Sequoia for storage systems, optimizing inventory management, and reducing order processing times by up to 25%.

Additionally, Amazon has integrated a technology that could revolutionize the world of e-commerce, which is drone delivery (amazon prime air).

Amazon prime air has launched in 2022, this service aims to deliver packages under 5 pounds in less than an hour using electric drones, which makes it ideal for fast deliveries. These drones are also capable of navigating through a variety of environments and weather conditions , including high temperature and rain. Amazon prime air was limited to two cities in the United States, but Amazon’s ambition was beyond limitation.

The company has unveiled the next generation drone delivery (MK30) and revealed that by the end of 2024, this amazon’s newest prime air model will officially take its first flight in europe, particularly in italy. This would transform the world of e-commerce to a whole new level in terms of customer experience and satisfaction.

In 2018, Amazon launched a technology that changed the traditional background of retail we know. Amazon GO. This groundbreaking technology combined the convenience of e-commerce and stores' physical experience. These technologies are implemented within all Amazon GO stores, enabling the cashierless and checkout free shopping experience. For example, the customer can visit the store, select his desired item and simply just walk out from the store. By machine learning algorithms, computer vision, and sensor fusion, the customer will be charged while exiting the store. Amazon's aim was always to enhance customer experience by leveraging new technologies and innovation approaches such as Amazon GO. Providing convenient, seamless , and personalized shopping journey for the customers.

Advantages and disadvantages of adopting new technologies to a business

Integrating new and emerging technologies such as AI recommendation systems, warehouse robotics, and drone delivery has significantly enhanced Amazon's competitiveness in the world of e-commerce. However, there are key advantages and disadvantages to be considered when adopting these kind of technologies :

Advantages:

1- The use of automation technologies like Titan and Sparrow mobile robotics in warehouse operations can reduce order processing times by up to 25%, minimize errors, and manage inventory more efficiently.

2- Using AI recommendation systems and cashierless system provided by Amazon GO improves the experience and increases customer satisfaction.

Disadvantages:

1- developing and applying technologies like robotics and drones can cost the company a lot of money and may require high financial investments.

2- since most companies rely on the existence of advanced technologies, this may reduce the need of humans. Especially in organizations like Amazon, where most of their operations can be done by automation and robotics, leading to massive social challenges.

Task 2

Customer Relationship Management (CRM) is an approach that involves using advanced technology to manage relationships with existing and potential customers. It helps businesses improve customer interactions, build loyalty, and achieve long term success.

In terms of e-commerce, the use of customer relationship management (CRM) helps the company to track customer behavior and purchasing history. Allowing the organization to personalize product recommendations and enhance shopping experience based on customer activities.

It is essential to implement a robust CRM for an e-commerce company like Amazon. Because it can improve customer service quality. Ensuring customer satisfaction and loyalty.

Amazon utilizes an AI powered recommendation engine. This system uses machine learning algorithms to analyze customer behavior, purchasing history, and customer's personal preferences. This engine allows Amazon to provide personalized recommendations. For example, when the customer purchases a particular item, the platform suggests related items that align with their purchasing preferences. This action is considered one of the key aspects of a successful, effective CRM.

Automation plays an important role in improving Amazon's CRM. The company has invested in developing mobile robotics such as Titan, Sparrow, Proteus, and Sequoia. And has made a revolution in its warehouse operations. These mobile robotics can manage inventory more efficiently, minimize the errors that could occur, and reduce processing times, which ensures that the customers receive their orders accurately and quickly.

Fast and reliable delivery is also worth mentioning since Amazon has developed and revealed its latest electric drone (MK30) as the newest delivery service model besides the traditional delivery services that also overcomes the majority of companies to maintain customer trust and satisfaction. Furthermore, Amazon GO uses automation technologies such as computer vision and sensor fusion to offer a cashierless shopping experience, which is the newest model that changed the face of traditional retail. Customers can simply select items and leave the store without the need for lines, and they will be charged once leaving the store. These actions are capable of enhancing customer convenience and loyalty.

After successfully integrating new and advanced technologies by Amazon, in my opinion, there is still a room for improvement. One of the emerging technologies that I would suggest implementing for a better CRM is Augmented Reality (AR) because this feature has the potential to transform the way customers interact with Amazon's platform and physical stores.

Augmented Reality (AR) is an interactive experience that combines the real world with computer generated content, displaying digital information like images, videos, or other 3D content onto a physical environment in real time. (source : wikipedia).

Benefits of Augmented Reality for Amazon's CRM

1- Augmented reality allows the customers to try on physical materials such as clothing, makeup, and accessories virtually. For example, the customer could check out how a piece of clothing would fit his/her size. Or to use their smartphones to see how a piece of furniture would look in their home. Implementing this technology can help Amazon personalize products and build a strong relationship with the customer, enhance customer experience, reduce hesitation of the customer, and maintain the loyalty and trust of the customer.

2- AR could also work with Amazon GO for guiding customers to specific product or section within the store, for example, when customer have some inquiries about a specific product or want to buy a product in a certain section, they could use AR app to locate their desired item to add to their shopping cart or helping them to access product details for any inquiries.

3- if a customer encounters any issue,by the use of AR, they could be connected with a support representative to guide them through the solution of their problems.

Task 3

Since Amazon has adopted and developed advanced technologies for their company, they have one of the strongest supply chain systems in the world by the help and use of these advanced technologies such as Artificial Intelligence (AI) and Machine Learning (ML), and AI robotics to ensure the best service including fast and cost effective deliveries to millions of customers around the world. In this task, I will mention the benefits of Amazon's supply chain system and the risks that could occur with it.

Supply chain in general is a system where products get distributed to customers. In terms of e-commerce, Supply chain is a number of procedures that requires an online store in order to operate. It involves activities such as warehousing, order fulfillment,and inventory management. It is very important for companies to have a strong supply chain because it can improve customer satisfaction and experience, it helps in managing inventory, it assists in order fulfillment, and it makes it easier to manage complex networks.

The benefits of an effective supply chain system

1- A strong supply chain system can directly impact customer's satisfaction when they receive their deliveries accurately and fast. For example, when Amazon decided to adopt and develop technologies and robots such as Titan, Sparrow, Proteus, and Sequoia, they made a revolution in warehousing operations because they reduced the processing order time by up to 25% which allowed them to earn customer trust and loyalty.

Amazon prime has provided two day shipping service on eligible purchases and exclusive items and benefits . This service has strengthened their supply chain and encouraged the customer to buy more.

2- Supply chains have become more effective by the use of AI and Machine Learning. These technologies have reduced the operational cost, minimized the errors, and improved resource allocation. For example, Amazon's AI and machine learning algorithms can predict product demand and manage inventory to avoid overstocking. This would impact the company to reduce the overall cost of operations and allows them to offer competitive pricing. Leading to achieve a high market position.

Consequences of supply chain management

1- the customer always expects the product to be available, fast delivered, and cost effective. Delayed deliveries and out of stock products can frustrate customers and can impact the company in terms of sales losing and negative reviews, leading to a potential loss of customer trust.

2- Overstocking unwanted products can cost the company because it requires more space, leading to increased unnecessary expenses and eventually losing profits.

Task 4

Online marketplaces have become one of the essentials of modern e-commerce.It connects millions of customers with billions of sellers. When it comes to companies, it is very important to understand how effective implementing new and emerging technologies into their online marketplaces can be to boost their revenues. In terms of Amazon, there is no doubt that this company has always been on top among strong competitors to adopt advanced technology to boost up their revenue. Emerging technologies in general can offer lots of benefits for businesses like personalising shopping experience through AI and machine learning, enhancing customer experience with chatbots to provide the best services,and providing a real world experience by Augmented Reality.

Amazon marketplace is one of the most successful online marketplaces in the world, it connects third party sellers with customers across the globe. Amazon has successfully used and developed advanced technologies to improve its online marketplace and conquer the world of e-commerce. These technologies were made to gain customer trust, enhance their shopping experience, and ensure their loyalty. Amazon was able to use technologies like AI and Machine learning for personalizing shopping experiences. For example, Amazon's AI algorithms suggest products based on customers browsing and purchasing history. Additionally, Amazon was also able to integrate automation technologies such as robotics to make a revolution in warehousing concept by developing Titan,Sparrow,Proteus,and sequoia for improving warehouse operations and reducing processing times. Implementing these technologies have helped them in enhancing customer satisfaction with maintaining their trust and loyalty.

Despite Amazon’s online marketplace being one of the most successful online marketplaces in the world, especially in implementing advanced technologies into their marketplace, there are lots of new and emerging technologies being produced to help such organizations like Amazon to further enhance their effectiveness. In my opinion, I would suggest adopting two emerging technologies like Blockchain and Augmented Reality (AR) to ensure safety, security, and the best shopping experience for customers.

But why adopting such technologies like Blockchain to Amazon ??

1- Blockchain technology can allow safe and secure transparent transactions, which would make a revolution in the Amazon marketplace. For example, most customers make purchases when they trust a product's authenticity, blockchain technology verifies the origin of the product to reduce the risks of fake goods. This would impact the revenue of the company by increasing the trust of the customer to make them repeat their purchases.

Why adopting and developing Augmented Reality (AR) ??

1- Currently, Amazon is expanding its use of Augmented Reality in its online marketplace to offer the best shopping experience for its customers by developing Amazon’s AR view, which is a feature that can allow the customers to explore and try out products in a real world environment before deciding to proceed with their purchase. Amazon has already spread this model in the furniture sections, which works by using the smartphones to place their 3d product to check their sizes and positions.

Amazon’s aim is to expand this technology in fashion and beauty by using and developing the feature (Virtual try on- try before you buy option). Allowing customers to see how items could look on them without the need for trying them physically. In my opinion, this would impact the company on boosting their revenues by reducing the hesitation of customers, leading to fewer returns and higher customer satisfaction.

Task 5

In the modern world of e-commerce, ensuring the security of customers' data and transactions is the most important aspect. There are so many kinds of Cybersecurity threats that negatively impact most companies such as, identity theft, data breaches, and unauthorized access. These threats are able to raise serious risks to such successful businesses like Amazon. In this task, we will explore the security measures that Amazon uses with the help of advanced technologies, how companies like Amazon protect their data from unauthorized access, and how Amazon ensures security and privacy of their customers.

Amazon uses advanced measures to protect customers' data. One of these security measures that Amazon uses to secure customers' data is encryption for data protection. Amazon uses advanced encryption protocols to protect customers' data. For example, Amazon uses high standard encryption algorithms to protect customers' sensitive information such as payment details, passwords, and personal data. These encryption algorithms ensure that the data remains secure and unreadable even if the data was intercepted.

Additionally, Amazon also utilizes Multi Factor Authentication to add an extra layer of security for users' accounts. Most companies use one time codes and passwords to be sent to users' devices as one of the multiple forms of verification. This would reduce the risk of unauthorized access to accounts.

Furthermore, Amazon benefits from the use of Artificial Intelligence (AI) and Machine Learning (ML) for detecting potential security threats. These systems have the ability to analyze large amounts of data, identifying potential fraud activities and anomalies, such as suspicious transactions and so many unusual login attempts.

Moreover, guaranteeing the security of payment information is essential for Amazon. The company ensures the security of payment transactions through robust systems and following data security standards. In addition, Amazon uses a system called (tokenization) for replacing sensitive payment data with unique tokens to reduce the risks of data exposure. These measures can protect the customers by securing their financial information and ensuring safe transactions. Leading to earn customers' trust and loyalty, making them visit the platform more frequently to buy more products.

Task 6

As mentioned earlier in the tasks, Amazon’s ambition is beyond limitation as the company’s mission statement is to be earth’s most customer centric company, where customers can find anything they need or might want to buy on Amazon’s online marketplace. (source: zooma.com). The adopting and implementing all the advanced technologies such as AI, robotics,machine learning, and blockchain . Amazon can guarantee long term success. Especially for the next 5 years.

As highlighted in tasks 1 - 5, the company has made good use of Automation and robotics, AI, and machine learning to make a revolution in its business models and to transform the company. Making it a top leader in the modern world of e-commerce.

Amazon has successfully integrated and used ai powered recommendation engines to provide the best service for the customers by suggesting products based on the browsing and purchasing history of the customers. The company emphasizes its commitment of enhancing its personalisation system for the customers by investing more in developing Augmented Reality AR. Augmented reality as mentioned earlier in the previous tasks allows the customers to virtually try out products like clothing or furniture for their homes to reduce the hesitation and increase the satisfaction of the customer . leading to an increase in the company's revenues.

In terms of security measures, this aspect has always been a top priority for the company to ensure the best customer experience and to protect customers' data and financial information by using high encryption protocols, powered by AI driven threat detection systems. This vision statement by amazon points out their focus on neverending development of these security measures to maintain the trust and the loyalty of the customers.

Additionally, Amazon's aim was not just focusing on security. Sustainability is also considered as a top priority for the company as they already started initiating electric vehicles and drones for next generation deliveries. For example, the company has introduced and launched the (MK30) electric drones that have the ability to navigate through various weather conditions including high temperatures and rain. This vision statement highlights how important these innovations could be for them and for the customers in terms of high quality services and how the company is dominating the modern world of e-commerce by its creative solutions for the ecosystem to place them as the top leaders for best supply chain management.

In my opinion, Amazon product strategy should focus more on the impact of using AI and machine learning for further enhancing their offers and to expand more in the top markets. For example, the company should invest more in developing Augmented reality because this technology has made a revolution in the way that customers interact with products. It is important to expand its use into fashion and beauty and not just be limited to furniture.

Additionally, it is very important to build a strong relationship with the customer to maintain Amazon's competitive position among the giants by implementing blockchain technology because this technology allows Amazon to verify product authenticity and provide

transparency for tracking orders. This would help in building trust between the customer and the company and reduces the risks of fake products.

Amazon should also prioritize the development of its chatbots to be able to handle more complex inquiries provided by customers. For example, integrating voice commerce using Amazon's voice assistant alexa can change the whole game in terms of providing a shopping experience where the customers’ are not required to input product details or names.

By implementing these technologies, Amazon can ensure a long term success for the next five years, and best services for its customers.

To reach their desired vision, which is to be the earth’s most customer centric company.

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